

Some Investment Aspects of Accumulation through Equities

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I

The terms of reference for this paper relate to systematic plans for saving or accumulation through common stocks. Such plans might include (a) a pension plan concentrating on equities, such as the CREF arrangement for college professors; (b) the very similar mechanics of the newly developing variable annuities; (c) systematic purchases of mutual-fund or closed-end investment company shares; and (d) an individual dollar-averaging plan, such as the monthly purchase program of the New York Stock Exchange.

The chairman has asked me to consider investment aspects of such plans in longer perspective and to give my views on the following questions: What results can be expected from them in the future as compared with the results either of the shorter-term past—i.e., since 1949—or of the longer-term past, going back into the last century and farther? How good will common stocks be as an inflation hedge in the future? Can dollar-averaging be counted on infallibly to produce satisfactory results? More specifically, can the much better performance of common stocks as against bonds be counted on to repeat itself in the next fifteen years?

I shall leave the chances and the effects of atomic war out of the following discussion, except for some observations regarding the indefinite continuance of the Cold War. Which of the two past periods will the future stock market resemble more closely—that of 1949–61 or that of, say, 1871–1961?

The latter comprises the ninety years for which we have common-stock indexes of earnings, dividends, and prices, as compiled first by the Cowles Commission and then continued by Standard & Poor's. In discussing equity experience in terms of the future behavior of these indexes, we are assuming that the various equity-accumulation plans within our purview are likely to show results, in the aggregate, approximating those of the S&P Composite or 500-Stock Index. It would seem easy enough to equal these average results; all that is needed is a representative diversification "across the board" and without that selectivity which is a watchword of today. But, paradoxically, if it is easy to equal the averages, it seems almost impossible for the average skilled investor to beat them.

We have very full information regarding the operations and achievements of the investment funds. For the 1949–60 period, as in the earlier years, they have not managed as a group to outperform the S&P 500-Stock Composite. It may be that professionally managed funds are too large a part of the total picture to be able to outperform the market as a whole; it may also be true, as I suspect, that certain weaknesses in their basic principles of stock selection tend to offset the superior training, intelligence, and effort that they bring to this task. But our main point is to establish that what happens to the stock market in general is going to happen to the typical or average accumulator of equities by any plan or under any auspices.

Let us attempt to summarize briefly the chief characteristics of the two market periods to which we are turning for clues to the future and among which we may have to choose. Table 1

Table 1*

A Picture of Stock-Market Performance, 1871–1960 and 1947–61

Period	Average Price	Average Earnings	Average P/E Ratio	Average Dividend	Average Yield (Percent)	Average Payout (Percent)	Annual Growth Rate**	
							Earnings (Percent)	Dividend (Percent)
1871–80	3.58	0.32	11.3×	0.21	6.0	67		
1881–90	5.00	0.32	15.6	0.24	4.7	75	–0.64	–0.66
1891–1900	4.65	0.30	15.5	0.19	4.0	64	–1.04	–2.23
1901–10	8.32	0.63	13.1	0.35	4.2	58	+6.91	+5.33
1911–20	8.62	0.86	10.0	0.50	5.8	58	+3.85	+3.94
1921–30	13.89	1.05	13.3	0.71	5.1	68	+2.84	+2.29
1931–40	11.55	0.68	17.0	0.78	5.1	85	–2.15	–0.23
1941–50	13.90	1.46	9.5	0.87	6.3	60	+10.60	+3.25
1951–60	39.20	3.00	13.1	1.63	4.2	54	+6.74	+5.90
1951	22.34	2.45	9.1	1.41	6.3	58		
1961(H)	72.20	3.10	23.2	1.97	2.7	64	+2.5	+3.5
1947–49	15.71	2.18	7.1	0.97	6.4	45		
1953–55	31.64	3.02	10.1	1.51	4.8	50	+5.7	+7.8
1959–61	59.70	3.24	18.3	1.91	3.2	58	+3.5	+5.9

* The above data are based largely on figures appearing in N. Molodovsky's article "Stock Values and Stock Prices," Financial Analysts Journal, May, 1960. These, in turn, are taken from the Cowles Commission book Common Stock Indexes for years before 1926 and from the spliced-on Standard & Poor's 500-Stock Composite Index for 1926 to date.

** The annual growth-rate figures are Molodovsky compilations covering successive twenty-one-year periods ending in 1890, 1900, etc. The lower growth-rate figures cover the ten years ended 1961, the six years ended 1953–55, and the twelve years ended 1959–61. The 1961 price is the high to December 7.

gives some data covering the three salient factors of earnings, dividends, and price behavior. We bring down the 1949–61 period into two six-year halves. The longer span can be handled in a variety of ways; we have decided to supply average computations for each of the nine decades between 1871 and 1961.

A cursory study of our table shows a number of striking differences between the exhibits of the two periods. Molodovsky has demonstrated that the overall gain to investors for the eighty-eight

years, 1871–1959, has averaged about 5% per annum in dividend return and 2½% per annum in price appreciation—both taken against annual market prices. The 2½% annual increase, compounded, in market price was closely paralleled by the annual rate of growth in both earnings and dividends.¹

But between 1947–49 and 1959–61 the growth rates for the various components have been quite diverse, and they have also varied sharply between the first and the second halves of the period. Thus we get very different indications of recent performance if we consider earnings rather than market price and if we consider the last six years rather than the last twelve.

The behavior of the stock market itself has been significantly different in the last twelve years from any previous period of equal length covered by our records. We have experienced what appears to be a single bull market, beginning at 13.55 for the composite index and rising to a current high of 72. The advance has been interrupted by three recessions, each on the order of some 20%. Under the usual terminology, these would be characterized as corrections or setbacks within a bull market.

As is well known, the long-term history of the stock market has been completely different. The picture, shown in Table 2, is one of a succession of bull and bear markets of varying duration and amplitude. Between 1899 and 1949 there were ten such well-defined cycles, thus averaging five years in length. (The longest period between peaks was ten years, from 1919 to 1929, and the shortest was two years, from 1899 to 1901 and from 1937 to 1939.) Most of the declines in the industrial averages were in the range of 40% to 50%; in the earlier years the composite average had somewhat smaller losses. The largest, of course, was the shocking fall of the Composite Index from 31.92 in 1929 to a low of only 4.40 in 1932, a loss of 86%. About the same proportionate decline was shown by the Dow Jones Industrial Average.

It should be pointed out also that the gyrations of past stock markets may not properly be viewed as taking place around a well-defined and persistent upward trend line, sloped at 2½% per annum. Both the price record and the earnings record disclose an irregular, rather than a regular, trend line. This should be evident from an inspection of our successive ten-year average figures between 1871 and 1950 and the Molodovsky growth rates appended thereto.

Table 2

Major Stock-Market Swings between 1871 and 1949

Year	Cowles-Standard 500-Composite			Dow Jones Industrial Average		
	High	Low	Percent Decline	High	Low	Percent Decline
1871		4.74				
1881	6.58					
1885		4.24	28			
1887	5.90					
1893		4.08	31			
1897					38.85	
1899				77.6		
1900					53.5	31
1901	8.50			78.3		
1903		6.26	26		43.2	45
1906	10.03			103		
1907		6.25	38		53	48
1909	10.30			100.5		
1914		7.35	29		53.2	47
1916	10.21			110.2		
1917/8		6.80	33		73.4	33
1919	9.51			119.6		
1921		6.45	32		63.9	47
1929	31.92			381		
1932		4.40	86		41.2	89
1937	18.68			197.4		
1938		8.50	55		99	50
1939	13.23			158		
1942		7.47	44		92.9	41
1946	19.25			212.5		
1949		13.55	30		161.2	24

The price-earnings ratio has also shown wide fluctuations. Since earnings have been even more unstable than prices—because of the recurrent business cycles—there has been a clear-cut tendency for the highest earnings multipliers to be established in years of depression, when profits tended toward the vanishing point. Our average figures by decades smooth out this type of variation; but they do not show the upward trend in price-earnings ratios that we might have

expected to accompany improvement in the underlying strength of our corporations and in the dependability of published earnings figures. Actually, the ratio of 6.2 times for 1949–50, at the beginning of the present bull market, was the lowest for any two-year period in our ninety-year history of stock prices and earnings. A devotee of the pendulum-swing theory of economic phenomena might well explain the current record-high multiplier as a reaction to the opposite extreme from the 1949–50 depreciation of the earnings dollar.

II

With these two quite diverse stock-market pictures before us, let us now inquire what are the respective arguments for considering the 1949–61 period as an integral continuation of the 1871–1949 span or for recognizing it as a new dispensation which will determine the character of the markets of the future. It is not too difficult for the student to fit the pattern of the recent market into that of the longer past. True, both the duration and the extent of its rise are already greater than those of any other.²

But that a pattern of many decades should establish new records of various kinds from time to time is only to be expected. A new record does not create a new pattern or character.

Is it possible, then, that we are living through a modified, but not essentially different, version of the experience of the 1920s? That there must be impressive differences goes without saying; for otherwise all of us would have been so struck by the incipient similarity as to make its continuation impossible. What may be more likely—speaking in abstract terms—is that the original differences have convinced us that the 1929 experience is irrelevant but that the similarities have been developing later, gradually, and so insidiously as to find us psychologically incapable of recognizing them. Let us attempt an enumeration of the major differences and resemblances between the current market and that of the 1920s, as they appear to this observer.

The two major internal differences relate to financial manipulation of various sorts and to excessive borrowing for speculation. The bull-market heights of 1929 were made possible by a huge wave of buying on thinner margins than are now permitted. Brokers' loans rose from \$2.769 million in 1926 to \$8.549 million in 1929, at which time they constituted about half of total member-bank loans. By contrast, the corresponding rise to date has been relatively much smaller. (Borrowings on smaller margins through other sources are no doubt significant at present but not sufficiently so to change the broad picture.) In the field of financial practices, the major abuses of the 1920s consisted of crass manipulation of stock prices by speculative pools and of corporate pyramiding through successive tiers of holding companies of various types. Both stock-market manipulation and corporate-structure manipulation have been greatly restricted by the S.E.C. legislation and by tighter stock-exchange supervision.

The amount that escapes detection is comparatively small, in my view. Although the various investigations now under way produce some startling exposés, whatever abuses now exist will not be found to have permeated the whole fabric of finance as was the case thirty years ago.

An exception to the above reassuring statement may have to be made in the field of new offerings of common stocks. Here I think a set of at least semi-manipulative practices has developed in handling so-called hot issues. The number of such offerings has been increasing steadily in the last two years, and their quality has been retrogressing at an equal rate. It is in this speculative area that I sense the closest parallel between the internal market

conditions of the late 1920s (and particularly of 1919) and those of today. Whether the new-issue financing of dubious merit will prove to be so heavy in aggregate dollars as ultimately to turn the market scales definitely downward, I shall not venture to guess. It is not impossible.

The widespread belief that we are in a new stock-market era, differing in its essential character from the bull-and-bear sequences of the past, rests on a number of claimed differences between then and now. These go well beyond the reforms in stock trading and in corporate financial practices. The case to justify the present unprecedented level of stock prices and earnings multipliers is essentially that which would justify the concept of a permanently changed character and future for the stock market. The safety and attractiveness of common-stock investment today is thought to be solidly grounded on a complex of favorable factors. Among them are (a) assured growth of population and GNP; (b) a rate of expansion more rapid than formerly, created by technological progress and the rivalry with Russia; (c) an assurance against major depressions provided by the government's new responsibility to prevent or quickly terminate them; (d) the public's recognition that common-stock investment is a necessary protection against continued inflation; and (e) the emergence of mutual funds, pension trusts, and other institutional investors as the chief source of demand and continuous support for common stocks.

Those who study the record of the 1920s will find that reasons similar to most, but not all, of the above were advanced to justify the ill-fated market rise of those years. The doctrine of "Common Stocks as (the Best) Long-Term Investments" emerged in 1924 and was made the cornerstone of the market's philosophy and its excesses. There was the same optimism about the future growth of the country and perhaps a better-founded confidence in the share of common-stock earnings in that growth. (The rate of return on invested capital was better maintained between 1922 and 1929 than between 1950 and 1961.) Old standards of value—particularly the once normal relationship between bond yields and common-stock yields—were thrown aside then as now, on the grounds that they had no relevance to the new economic climate. There was great confidence, also, in the future stability of business and its immunity from severe depressions. This was founded on the idea that scientific management, careful control of inventories, the absence of inflation, and other factors would help our business leaders avoid the costly mistakes of the past.

In my view, there are three major differences between the economic realities of the 1920s and the present. The first relates to the inflation factor, the second to the Cold War, and the third to the role of government in business. The bull market of the 1920s ran its course without the aid of commodity-price inflation; the market rise since 1949 has been accompanied by an irregular, but virtually continuous, advance in wholesale and consumer prices. It is difficult to say whether the investor's current emphasis on future inflation possibilities should be considered primarily as a recognition on his part of objective fact or rather as a strong subjective reaction to an element that is by no means new to the financial scene. We had more inflation in wholesale prices from 1900 to 1910 than from 1950 to 1960; the rise from 1900 to 1920 also exceeded that from 1940 to 1960 (the equivalent of from 36 to 100 versus from 51 to 120). Most of us believe that inflation is the path of least resistance for governments, labor leaders, and business heads and that hence it will be followed. But the record of the past will not help us much to determine what the amount of inflation will be over future decades, whether its course will be regular or interspersed with sharp deflations, as in 1921 and 1932, and whether investors will remain as inflation-conscious in the future as they are today. The reaction to inflation, like almost every other investment and speculative attitude, seems to be more the result of the stock market's behavior than the cause of it.

My view of the effects of the Cold War on common-stock values is quite a personal one, not shared by many, I am sure. In the first place, I think that it has contributed a good deal to the business expansion and relative stability of the past decade. But, in a contrary sense, I cannot see how the kind of Cold War we are now living through can continue throughout "our lifetime and that of our children." Sometime within the present decade, a way will have to be found to terminate the Cold War, or it will be transformed into large-scale hostilities, with all their nuclear implications. If our prosperity since 1949 has, in fact, rested rather heavily on our defense expenditures and if, in truth, we must fairly soon have either no war or nuclear war in place of Cold War, then today's international situation cannot be termed more favorable for common stocks than the cloudless one of 1929.

The government's commitment to prevent large-scale unemployment and serious depressions is both a new factor and one of major importance. The most logical reason for expecting a different kind of stock-market cycle in the future than in the long-term past would appear to be by analogy with the business cycle. The record since 1949 strongly supports this thesis. The new material on "Business Cycle Developments," now available monthly, shows four periods of business contraction since 1948—in 1949, 1953–54, 1957–58, and 1960. All these were very moderate, as compared with the sharp recession of 1937–38 and the major depressions after 1919 and 1929. The three declines of about 20% each in the stock averages since 1950 appear to correspond fairly well to the three setbacks of about 10% in the index of industrial production. If we have now entered a new era that excludes old-time business depressions, it seems reasonable to deduce that we are also in a new era that precludes old-fashioned bear markets.

III

Both my analysis and my instinct warn me that there may be a catch in this plausible and reassuring parallel. If the recent picture had been one of the stock market's advancing in step with the national product and in close proportion with it also, then the observer might conclude—somewhat to his amazement—that not only has the economy been reformed but human nature as well. But here the facts part company with the hypothesis. The stock-market level has not been governed primarily by the level of business but rather by the development of new investment theories and attitudes and by a typical growth of speculative interest and activity. Some of the old financial abuses that characterized former bull markets have, indeed, been virtually eliminated. But some have again raised their heads, and some new ones have appeared and are spreading apace. These are in the areas of corporate reporting, corporate financing, the quality of the enterprises offered for public sale, and the ways in which new issues of common stocks are offered and subsequently traded.

Equally important and dangerous, in my eyes, is the ready acceptance by security analysts of the going market levels and earnings-multipliers as the proper standard of value and of comparison for any issue under study. The new analytical concepts of growth-stock valuation, of "cash flow," of the desirability of tax-free dividends from companies which are triumphantly able to report earnings deficits—all have enough plausibility and lack of inner discipline to lead both investors and speculators far astray. In sum, the new investment theories and techniques remind me very much of 1928–29, and the outpouring of common-stock issues of secondary and lower-degree enterprises reminds me equally of 1919. If the relative stability of general business and corporate profits produces an unlimited enthusiasm and demand for common stocks, then it must eventually produce instability in stock prices. We have already seen the working of this paradox in the area of growth stocks. The price of a successful and promising concern such as Texas Instruments can be driven up so high by speculative emphasis on its

prospects that the ensuing reaction has cut the price in half—with no change in the underlying worth of the business. Examples of this sort are now numerous. Conceivably, this behavior of issues in the growth-stock class may give us a preview of the ultimate behavior of the general market—as represented by comprehensive averages—if common-stock investment becomes essentially identical with common-stock speculation. In that case the stock market will have a life-cycle of its own, quite independent of the business cycle. The market cycle will once more prove to be the human-nature cycle; its economic background will have changed but not its basic character or the consequences of its character.

These arguments against a new character for the stock market are not necessarily arguments that the present levels are too high, although they certainly would be adjudged so by older standards. Conceivably and even probably, new factors in the economic figure have moved upward the central value of the average dollar of corporate earnings and justify a more favorable relationship than heretofore between stock yields and bond yields. This would certainly be true if the general business picture can be counted on to continue indefinitely the relative immunity to depression it has shown since as far back as 1941. What we are concerned with here is not the future central value of the stock market but rather the amplitude and the consequences of possible future variations around this value.

To soften a possible charge of old-fogysm and prejudice against new standards of value, may I take this paragraph to show how the recent record level of stock prices may be justified by some not implausible calculations. Let us assume that the investor wants an overall return of $7\frac{1}{2}\%$ annually, as a composite of dividend income and average market appreciation. (This $7\frac{1}{2}\%$ target is itself taken from the long-term record of dividend yields and price advances; it seems reasonable as a guide to the future.) Assume, next, that earnings and dividends will grow in the indefinite future at the annual rate of $4\frac{1}{2}\%$, which appears to be the projection for this decade. Then the investor should be satisfied with a 3% dividend return. This would justify a current level of 65 for the S&P Composite, only 10% below the recent high. A small adjustment here or there would put us over the top.

It is by no means impossible to assume a permanent growth rate of $4\frac{1}{2}\%$; we have been told that we must increase our GNP faster than this or lose out in the race with Russia. The basic objection is that it is only an assumption, that the experience of the longer past puts the figure rather at $2\frac{1}{2}\%$ and that the difference between $4\frac{1}{2}\%$ and $2\frac{1}{2}\%$ in this calculation means the difference between 65 and 39 for the value of the S&P Composite. My experience leads me to predict that the action of the market will govern the investor's choice as to probable future growth rates rather than vice versa.

This completes my case for and against a new era and character for the stock market. If the market since 1949 foreshadows the stock markets of the future, the investment aspects of equity accumulation are unbelievably favorable. All that will be needed will be the funds to buy a representative assortment of common stocks and a little patience to sit through periods of mild reaction.

The annually compounded rate of overall return from the 500-Stock Composite has been about 13%—curiously enough, this has been about the same as the average from a selected list of growth stocks.³ A much lower annual rate, without severe interruptions, will prove amply rewarding.

IV

But if, as I deem more likely, the fundamental character of the stock market must be as unchanging as that of human nature, then the accumulator of equities who starts today is faced with quite a different prospect. The new appearance of the variable annuity suggests a broad analogy with the 1920s. It was during that bull market that investment trusts had their first important development in this country. Most of the arguments in their favor were the same as those now used for the sale of mutual-fund shares and for equity accumulation in general. The collapse of 1929 resulted in a severe and protracted setback for the investment-trust movement, as part of a widespread loss of faith in common stocks generally. It is true that many—although perhaps less than half—managed to survive the bitter subsequent experience and to reestablish themselves more firmly than before in public esteem. Furthermore, the principle of dollar-cost averaging—which is the most systematic of the equity-accumulation techniques—was able to vindicate itself in the end, after perhaps twenty years of unsatisfactory-to-mediocre results.⁴

The computations made of theoretical dollar-averaging experience in the past embolden us to predict that such a policy will pay off ultimately, regardless of when it is begun, provided that it is adhered to conscientiously and courageously under all intervening conditions. This is by no means a minor proviso. It presupposes that the dollar-cost-averager will be a different sort of person from the rest of us, that he will not be subject to the alternations of exhilaration and deep gloom that have accompanied the gyrations of the stock market for generations past. This, I greatly doubt—particularly because most of the dollar cost-averagers we are speaking of will be typical members of the public who have been persuaded to embark on an equity-accumulation program by the arts of salesmanship now so highly developed in the mutual-fund field.

Let me return once again to the problem of the proper perspective for viewing the character of the stock market and the investment aspects of equity accumulation. At the outset I presented a statistical comparison of the market's behavior over the last twelve years and over the last ninety years. But our knowledge of stock-market behavior goes back a good deal more than ninety years—a full two centuries and a half, in fact, to the inception of the South Sea Company in 1711. In our first edition of *Security Analysis*, published in 1934, we characterized the stock-market madness of the 1920s as a repetition or rerun of the famous South Sea Bubble. By comparison, the behavior of our present market appears more rational, dignified, and reassuring. No one today, not even ingrained conservatives (like this speaker), expects consequences to this market and the economy even faintly resembling the catastrophe of 1929–32. Yet I have a feeling that the financial world has become too complacent about the future, too confident of the invulnerability of common stocks as a whole to a drastic change in their fortunes.

A great corporation can withstand great vicissitudes; the same is true of great institutions, among which not the least important is common-stock investment and equity accumulation over a span of time. But a bull market has never become a financial institution, and I have great doubts whether this attractive development is an admissible possibility, when the frailty of human nature is taken into account.

My own inward picture of the present stock market is that of an institution cut adrift from old standards of value without having found dependable new standards. (In this respect present-day investment may be in somewhat the same position as present-day painting.) The market may either return to the old measures of central value or—as is perhaps more probable—eventually establish a new and more liberal basis for evaluating equities.

If the first happens, common stocks will prove highly disappointing over a long period for many accumulators of equities. If the newer and higher value levels are to be established on a sound basis, I envisage this working out by a process of trial and error, covering an unpredictable period of time and a number of pendulum swings of unforeseeable magnitude. I do not know whether bonds will do better than stocks over the next fifteen years, but I do know that the people behind College Retirement Equity Fund (CREF) are eminently wise in insisting that its beneficiaries have at least an equal dollar stake in bond as in stock investment.

Notes

1. See N. Molodovsky, "Stock Values and Stock Prices," *Financial Analysts Journal*, March, 1960.
2. These statements refer to the S&P Composite Index. The percentage rise in the industrials was somewhat larger from 1921 to 1929 than from 1949 to date, because it started from a relatively lower level.
3. See, for example, J. F. Bohmfalk, Jr., "The Growth Stock Philosophy," *Financial Analysts Journal*, November, 1960, Table A.
4. For calculated results of dollar-cost-averaging results for ten-year periods ending from 1929 through 1952, see Lucile Tomlinson, *Practical Formulas for Successful Investing* (New York: Wilfred Funk, 1953), Table 3, p. 62